

UNCHANGED BY THE SWAP

robodog_control_node
loop, safety, gait, estimator

robodog_web_server
GUI, protocol

robodog_camera_node
publishes RGB-D

RViz2 / rosbag2
same topics

JointBackend

abstract: configure/enable/read/write/step

KinematicBackend
no physics, runs in CI

MujocoBackend
1 ms command delay

2 x CAN @ 1 Mbit/s
python-can, 400 Hz

RoboStride02Backend
RS02 frame codec

pure, 29 unit tests

CameraBackend

abstract: configure/capture/intrinsics

SimCameraBackend
range noise

NuwaHP60CBackend
UVC / vendor SDK

direction +1 / -1
offset_rad
bus, motor_id
Per joint mapping
coordinates differ from
canonical joint coordinates.
Set by
calibrate_joint